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Hiring Manager
Engineering Team
NXTGEN Industries
121 King Street
Melbourne, VIC 3000

Dear Hiring Manager,

RE: NXTGEN Intern Challenge, 2026 Intake

I am writing to apply for the NXTGEN Intern Challenge 2026 Intake at NXTGEN Industries, advertised on your careers page. I am a third-year Bachelor of Software Engineering (Honours) student at Deakin University, and the Challenge appeals to me because it asks interns to solve one real robotics problem together, across every discipline, rather than in isolation.

I hold a High Distinction average and expect to complete my degree in 2028, with marks of 100 in Backend Development and in Data Structures and Algorithms, and 93 in Embedded Systems. The skill I would bring first is integrating software with physical hardware. This year I built a sensor-driven robot on an Arduino and a Raspberry Pi, communicating over MQTT, and then wrote a separate .NET backend that controls it over Wi-Fi. Making the two work together meant debugging problems that sat between disciplines, including a power fault I eventually isolated from the robot's own diagnostic logs. The second is treating field data with suspicion. Both systems check what they are told before acting on it or recording it, because a reading from the field is not the same as the truth.

The third is testing claims rather than assuming them. When I load-tested my backend against a simulated fleet, it failed completely at scale. I identified twelve bottlenecks and race conditions, fixed them, and proved the result with 54 automated tests and 80 benchmark runs before reporting any improvement. Outside study, I volunteer as IT support for the Boroondara Stroke Support Group, where I trained other volunteers to take over a monthly publishing task, so the work no longer depended on me.

NXTGEN stands out to me because it carries data all the way through to action. Many companies stop at analytics or at a report, while your Inspect, Autonomy and Flow products connect defect detection to robots that work in confined spaces, at height and in hazardous environments, and to the workflows that act on what they find. Projects such as rail and train undercarriage inspection show how much depends on that data being trustworthy, which is the part of robotics I find most interesting. I also value that your platform stays embedded in a client's operations instead of being delivered and left, because the systems I have learned most from are the ones I kept improving after they first worked. Within the Challenge, I would contribute most on software and hardware integration, and I am keen to develop in control systems, sensing and printed circuit board design, which I have not yet worked on.

I would bring strong software fundamentals, hands-on hardware experience and a habit of verifying results, together with a genuine interest in the work NXTGEN does. I am available for a continuous block of three to six months and for regular in-office work in Melbourne. My resume and project portfolio are attached. I would welcome the opportunity to discuss my application at interview. Thank you for your consideration.

Kind regards,

Manit Khera